

Learning rate

 $1 \times 10^{-4}$  $3 \times 10^{-4}$  $5 \times 10^{-4}$  $r=4, \alpha=4$  $r=4, \alpha=8$  $r=8, \alpha=8$  $r=8, \alpha=16$  $r=16, \alpha=16$  $r=16, \alpha=32$  $r=32, \alpha=32$  $r=32, \alpha=64$  $r=64, \alpha=64$  $r=64, \alpha=128$  $(r, \alpha)$ 

0.933

0.942

0.943

0.948

0.947

0.951

0.951

0.956

0.954

0.959

0.949

0.955

0.952

0.956

0.959

0.959

0.961

0.957

0.959

0.956

0.956

0.959

0.958

0.958

0.959

0.959

0.958

0.958

0.958

0.954

0.95

0.90

0.85

0.80

Mean f1